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Engineering Analysis and Design 1A (FEBE1002A)

Modular Drip Irrigation Planter Box

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Abstract

A request was made to improve the standard of life in Pu Ngaol, Cambodia by cost effective and sustainable means. Food stability was found to be lacking in the village and as a result the objective for this study was to improve their access to food on a sustainable basis. A design was developed that ensures the people of Pu Ngaol will have a more secure source of food and nourishment that will be able to withstand the everchanging water security level in the region. Through research, it was decided that the optimum solution was a water effective modular drip irrigation planter box. The resultant design allowed for the maximum use of available water for subsistence farming, ensuring optimum use of water to grow food.

1. Introduction

1.1 Context

The rural Cambodian village of Pu Ngaol village lies within a protected national park in Mondulkiri province, about 400km northeast of Cambodia's capital Phnom Penh. In Pu Ngaol there are significant problems in relation to food access, agricultural techniques, and water access. In this village of roughly 400 people, most of the economy relies on the agricultural production of rice paddies and the collection of resin from forests. Pu Ngaol faces numerous formidable challenges in supplying this agriculture with a sustainable water supply which is further exasperated by the lack of a suitable and efficient planting and cultivating technique. For the local farmers, who rely on agriculture as their main source of revenue, water is vital and good soil and coherent agricultural knowledge, and techniques are in short supply. Due to factors like population growth, drastic seasonal variations in climate and a shortage of water supplies this results in a community that is paralyzed economically and health-wise.¹

1.2 Problem statement

We were presented with the problem of improving the living standard of the people living in Pu Ngaol, Cambodia in some way. We have decided to focus on the issue of the lack of available food for the people of the village, due to lack of water to use in subsistence farming. In this region of Cambodia there is not a suitable agricultural method in Pu to combat the harsh temperatures and inconsistent weather conditions. There is very little rainfall in the dry seasons of the year thus leading to not enough water for the crops². Furthermore, there is too much water during the wet season, resulting in increased flooding of rice fields and other crops. There is little infrastructure in place that is able to cope with the destruction of cultivated areas, collection of rainwater and ways of increasing crop output. This is due to the lack of knowledge of farming techniques and initial capital needed to set up infrastructure able to deal with these problems.

In the past, farmers have resorted to using borehole water to water subsistence crops. This water contains mud, high levels of calcium carbonate and iron. Whilst small scale farming is not entirely hindered by this, these conditions result in a lower yield and quality of the crops watered by this water. This instability of crop abundance causes a decline in the community's general health, as people are not gaining the necessary nutrients from their food as they need.

In order to combat these problems, we will design a way for families to be able to have a reliable way to grow crops in order to maintain healthy lifestyles. Our design is required to be cost effective and not demanding of the environment in Pu Ngaol.

We have thus decided to design a water efficient modular drip irrigation planter box. We propose to create this planter box using as cost effective and environmentally low impact materials as we can. The design will make use of grey water and rainwater.

1.3 Justification for design solution

The main design proposal is a Modular Drip Irrigation Planter Box. This design will be referred to as the “Planter Box” in the rest of the report. This design is a planter box, of dimensions 5 by 3 meters and 0.55 meters high. In this box, soil will be deposited, and a specific selection of leafy vegetables will be grown such as carrots, potatoes, and tomatoes. The main function of the system is the fact that it uses a drip irrigation system. The concept of drip irrigation will be discussed later on. This system allows for massive water savings to occur on a small-holdings subsistence scale. Furthermore, vegetable crop quality and quantity will increase. This helps solve the ever-present food demand in Pu Ngaol.

There is a significant water insecurity in Pu Ngaol as discussed. By using a system that is known to be highly water efficient, (up to 90%)³ it puts relief on the water security of the village, ensuring that farmers can effectively conserve water in comparison to the large water use in rice farming. Some more benefits include improving food security, food and nutrition diversity and saving land use. Moreover, with the considerable influence of global warming on Cambodia⁴, a water-efficient system that is able to cope with incompatible climatic conditions will prove invaluable for the village. A further benefit of this design is that the system uses a gravity fed system (system works by force of gravity only), which is a significant way of mitigating the disadvantage due to lack of electricity. The planter Box design aims to be a simple, affordable, expandable crop growing system that can be effectively used by the village. Drip irrigation has been implemented in some regions of Cambodia before and it has shown to be very effective. A study was done which showed how drip irrigation increased the yields and development of vegetable production during the dry season in the country.⁵

2. Drip Irrigation

2.1 What is drip irrigation?

The proposed design solution of this report employs a groundbreaking irrigation method for agriculture known as drip irrigation. The core principle of drip irrigation involves delivering a specific and calculated flow of water directly to the root zone using a device called an emitter⁶. This method is designed to minimize evaporation and conserve a significant amount of water. It also ensures that each plant receives exactly the right amount of water it needs for optimal growth. This technique enables farmers to produce higher yields while saving water, fertilizer, energy, and crop protection products⁷. Drip irrigation provides an alternative to more water-intensive irrigation systems like furrow, flooding, and overhead sprinkler irrigation, which are difficult to set up and require larger areas of farmland. Drip irrigation systems distribute water

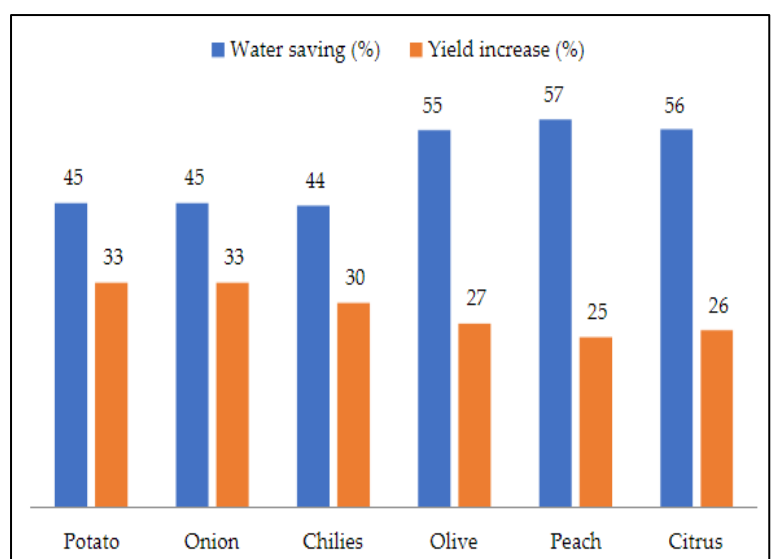


Figure 1 Graph showing the percent increase in water saving and yield when a drip irrigation system was used in place of a furrow system. From the study conducted in Pakistan.

through a network of valves, pipes, tubing, and emitters. When well-designed, a drip irrigation system can be much more efficient than these other types of irrigation systems. These unmatched benefits made drip irrigation the ideal choice for this design. Drip irrigation can reduce the total water consumption of some farms as much as 60% to 90% depending on the type of drip irrigation system used, as well as soil, moisture, plant, and climate types⁸. Furthermore, drip-irrigation has optimized the use of farming land, due to the exact nature of the placement of crops in such a system⁹. This means in order to generate arable farmland, far less land is required for drip-irrigation systems in comparison to other types. Multiple studies and research papers have shown that in a world under the influence after global climate change, drip irrigation has come to the limelight as a method that can conserve water resources unlike any other techniques. A study in Punjab, Pakistan showed massive water savings for the crops of farmers on small holdings when a simple drip irrigation system was used in place of a furrow irrigation system¹⁰. This study also showed a significant reduction in the costs in the construction and maintenance of a irrigation system when a farmer constructs a low-pressure drip irrigation system rather than a furrow or overhead one.

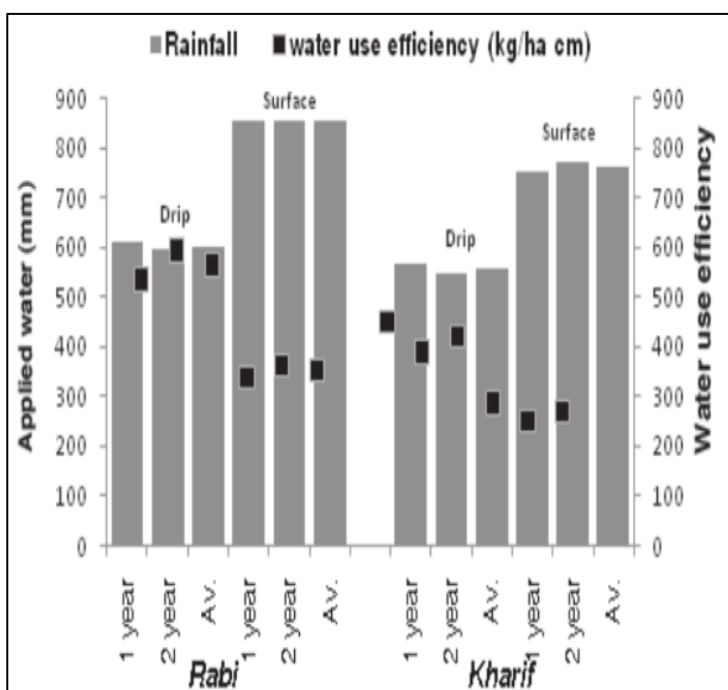


Figure 2: Graph showing the difference in water savings from surface and drip irrigation in farming onion types over 2 years.[12]

A study in China¹¹ conducted a global meta-analysis to evaluate the effectiveness of drip irrigation. The research indicated that during drought conditions, drip irrigation can save water and produce higher crop yields compared to flooding, border, furrow, sprinkler, and micro-sprinkler irrigation methods. Specifically, when the amount of drip irrigation is increased to 100-120%, crop yields are boosted by 28.92%, 14.55%, 8.03%, 2.32%, and 5.17%, respectively, over these other irrigation systems. In scenarios with ample water resources, increasing the amount of drip irrigation further enhances crop yields. Additionally, the researchers found that drip irrigation helps reduce fertilizer leaching and soil salinity.

Furthermore, drip irrigation has been shown to reduce the possibility of plant diseases occurring. This is due to the flow of water being slower and occurring very close to the soil. Drip irrigation prevents the problems that occurs when water applied through irrigation is in excess or deficient it results in fungal and bacterial diseases which reduces the crop quality and quantity.¹³

2.2 The disadvantages of drip irrigation

The main pitfalls of drip irrigation lie in its initial capital investment. The many components, devices, expertise, and maintenance needed in order to setup can be rather expensive. The emitters, filters and pipes are generally expensive and require significant planning in order to cope with the price. A level of knowledge must exist in order to create a suitable system in the first place, this means having professionals on hand. The maintenance of emitters is also labour-intensive, as they are constantly clogged with sediments and may be chewed up by rodents, thus more effort must be made to constantly repair the pipes and sub-pipes. Also, in terms of environmental sustainability, these systems can produce plastic waste as many of the components make use of plastic materials, i.e. the drip tape,

emitters etc.¹⁴ The discussion of the design solution will try to address these challenges along with the challenges associated with the specific setting of the problem itself.

2.3 Types of drip irrigation

There are two main drip irrigation types: sub-surface drip irrigation and surface drip irrigation.

Sub-surface drip irrigation consists of lateral tubes are under the soil surface. Narrow plastic tubes are buried in the soil at a certain depth depending on the crop grown, being deep enough so as not to interfere with the normal growth of the roots and to allow enough space for the water to infiltrate¹⁵. This approach reduces water waste and soil erosion while effectively controlling weed growth by keeping the soil surface dry. However, subsurface systems are more expensive to install due to the need for specialized equipment and can be damaged by roots growing into them, which can damage the system. Additionally, regular maintenance is essential to prevent clogging and ensure efficient operation.

Surface drip irrigation is a method where the soil is irrigated on the surface of the plantation, resulting in more water emitters used over a much larger area. This method conserves water by reducing evaporation and runoff, minimizes soil erosion, and requires low energy as it is often gravity-fed. However, the exposed tubing can be easily damaged, wind can disrupt the water flow, and it may promote weed growth between crop rows. The choice between these methods depends on budget, maintenance capacity, and specific farm requirements.¹⁶

A study done by researchers in Saudi Arabia¹⁷ found that when comparing the two systems in use of farming date palms in a desert environment , the amount of water applied to the subsurface-drip irrigation was 30 % less than the quantity of water applied to normal drip irrigation , where the average monthly water depth applied to NDI was 194 mm compared to SDI (136 mm) which is a difference of 58

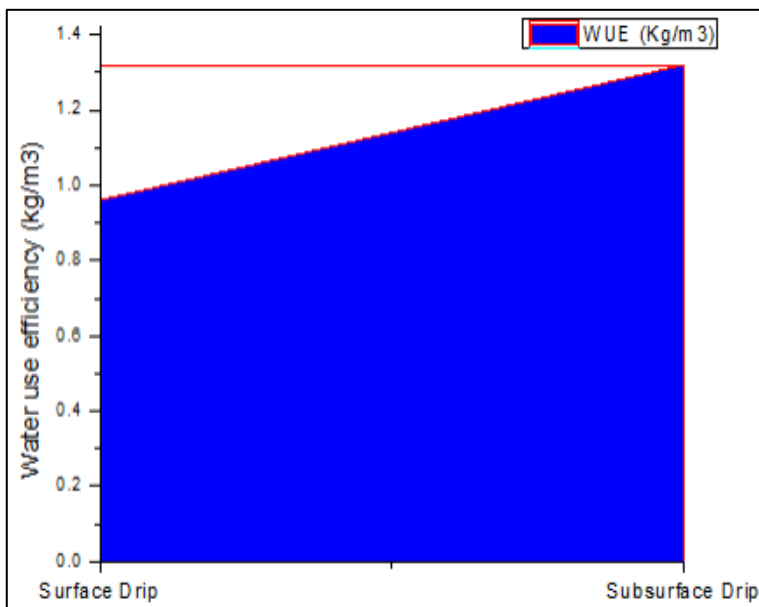


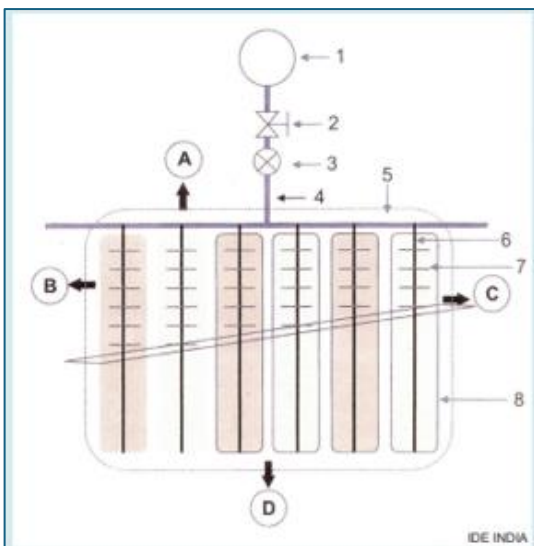
Figure 3 Graph showing the difference in water use efficiency from sub-surface and surface irrigation in farming date palm plants over an agricultural period year⁷

mm. A comparison of water use efficiency between both irrigation methods showed that water use efficiency was increased by 27% more in case of SDI (1.32 kg m-3) compared to DI (0.96 kg m-3). However these two systems had very similar outputs in terms of the quality and quantity of the fruit produced. Furthermore, in this study they showed that that while the initial economic cost of the subsurface drip was much more expensive, overall, over a longer period more money was saved on watering costs in comparison to surface drip farming.

2.4 The components of drip irrigation farming

Drip irrigation farming is unique in the sense that there are many ways and layouts to generate a specific design that implements the concept of drip irrigation, this is all depends on the quality and amount of water available, type of arable soil used, the climate of the environment, the total area of land needed

to be irrigated, the vegetables used, the level of soil moisture etc. Thus, the most crucial components are discussed.



1) Water Access and Storage: A reliable, clean, and sufficient water supply is crucial. This water can come from rainfall, standard water connections, boreholes, or rivers. The collected water must then be stored in an appropriate container based on the system's requirements.

(2) Control Heads: These are equipped with valves to regulate water flow and pressure throughout the system. They typically include filters, such as screen filters or graded sand filters, to remove debris. Some control heads may also feature a tank for fertilizers or nutrients ¹⁸.

(3) Water Pump: Drip irrigation systems operate under low pressure, so gravity can also be used in gravity-fed systems to pressurize the drip lines and pump water around the system.

(4, 5, 6) Main Distribution System: This consists of the mainline pipes, sub-main pipes, and lateral pipes. The distribution system is made of PVC pipes or any tubing large enough in diameter to deliver the required amount of water to the field at the correct pressure.

(7) Emitters (or Drippers): These devices control the water flow from the irrigation system to the plants. Their placement depends on the type of crop and soil. For row crops, closely spaced emitters can moisten a continuous strip of soil. Emitters deliver a steady, specific amount of water consistently, despite pressure variations, and are resistant to clogging.

Miscellaneous Parts: These include goof plugs to seal unnecessary holes in the mainline tubing, valves to control different sections, flush valves to clear debris and prevent clogging, and connectors and adapters to link different types of pipes.

2.5 Evaluating the water tank and water supply needed

Evaluating the water tank and the water supply needed is a crucial step in ensuring the efficiency and success of the modular drip planter box system. For vegetables such as tomatoes, potatoes, and carrots, which are designated for the planter box, a precise amount of water is necessary to promote optimal growth. These vegetables require careful estimation of water volume due to their classification in the "leafy greens" category, which typically demands consistent and specific hydration levels.

The water requirements for these vegetables fluctuate based on several factors, including seasonal variations and the specific growth stages of the crop's peaks. This variability necessitates a flexible water supply system capable of adjusting to the plants' changing needs throughout their growth cycle.

In Cambodia, the climate allows for the cultivation of these vegetables for most of the year, but it also poses unique challenges. The country experiences distinct wet and dry seasons, with the wet season providing abundant natural water sources through rainfall, while the dry season demands more reliance on stored water. Consequently, the water tank must be adequately sized to store sufficient water to sustain the crops during periods of low rainfall. To accurately determine the water tank capacity, it is essential to calculate the total water consumption for the entire planting area. This involves considering the daily water requirements of each vegetable type and multiplying it by the number of plants and the duration of the dry season. For instance, tomatoes generally require about 0.8 to 1.2 inches of water per

week, while potatoes need approximately 1 to 1.5 inches, and carrots about 1 inch. These figures need to be adjusted according to local environmental conditions and the specific irrigation efficiency of the drip system.

The design of the water tank should ensure it can accommodate the peak water usage periods without running dry. This might involve integrating multiple tanks or a larger central tank, with consideration for future scalability of the planter box system. Furthermore, incorporating rainwater harvesting systems can supplement the water supply during the rainy season, reducing the dependency on external water sources and promoting sustainability.

2.6 Filter design

In order for the water emitters to function at an optimal level, there needs to be a filter system in place to ensure that they do not become clogged by minerals and debris in the water. Through research, three of the most effective filters for this purpose are screen filters, media filters and disc filters.¹⁹

Screen filters allow for surface filtration, which removes particles from the surface of water. The screens used are often made of metal or plastic. The filter works through a coarse outer screen and a finer inner screen to ensure proper filtration of water. Whilst a screen filter has a lower cost for flow and requires a smaller area for installation, it only filters water through a singular layer of mesh and do not have a depth filtration as the other filter options.²⁰

Screen filters are mainly suggested to be used with high quality water, making this filter option not very optimal for the planter box design, as water quality in Pu Ngaol is poor. The filter would need to be backwashed very regularly, and this is not an option as water is scarce.

Disc filters are a combination of media and screen filters. They consist of a series of grooved rings, with each groove crossing over the others all in different directions, creating a three-dimensional connection of pathways in a cylindrical filter. Varying the sizes of the grooves allows for a different level of filtration to occur.

Disc filters require automatic flushing of internal jets to ensure optimal filtration. Disc filters are not ideal as they require a higher maintenance than media or screen filters, as there are more moving parts required to be dismantled during cleaning of clogs.

Media filters, such as sand filters, provide the highest degree of filtration and are the optimal choice when dirty water is being used, which accurately suites the water quality used in this design. These filters use the filter action of silica and granite, usually between 0.75 to 1.5mm in diameter. Media filters are well suited for drip irrigation due to their ability to trap large amounts of fine particles present in water.

Media filters can be upgraded by increasing the number of media levels in the filter, which in turn increases the number of particles trapped in the filter. Whilst media filters do need to be cleaned so that hydraulic rate is reestablished, this process is much easier than backwashing. Simply, the top 5 cm of sand is removed from the filter, which can then be replaced. Media filters are the most ideal filtration method, as it is the most cost-effective solution, and work with a gravity system, allowing there to be no need for a mechanical or electrical pump. Media filters also filter out the largest quantity of particles of the three proposed filters.

For the drip irrigation planter box design, a media filter has been deemed the best solution to filter the incoming water to prevent blockage of the water emitters in the irrigation system. More specifically, a slow sand filter design has been selected²¹. This allows for the most simple and effective technique for purifying the water running into the irrigation system. The filter consists of three different layers

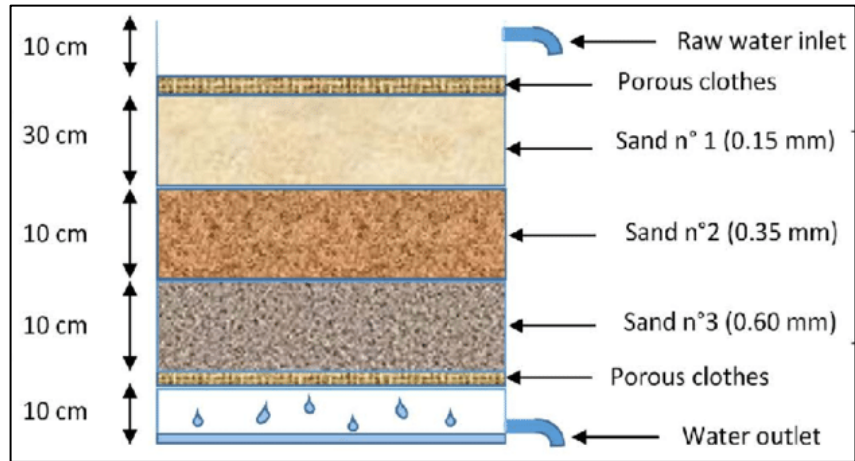


Figure 4: Slow sand filter design.²²

of media inside of the filter box. The layers are the underdrainage system, gravel, and sand. Sand is readily available within Pu Ngaol, making this filter optimal.

2.7 Evaluating drip rates of the drip irrigation emitters

In the precision process of a drip irrigation system, it is crucial that the correct delivery rate is determined, and the correct dripper spacing requirements for the most optimal conditions. In the proposed “Modular Planter Box” design, this had to be considered in two aspects: What is the most effective and simple way to implement the correct drip rates and drip spacing into the design and how does the system implement an effective emitter design based on the water, soil, temperature needs on the vegetables and also considering the evapotranspiration rates, (how quickly is water evaporated from the plants).

The first factor to consider when designing the main drip emitter system is the type of soil of the farm.

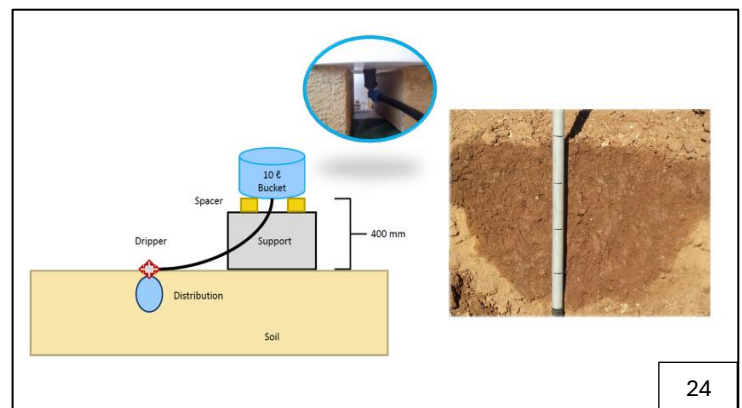
2.7.1 Soil's influence on the drip emitter design

Soil infiltrability: this is determined by the proportions of sand, silt, and clay, and the rate at which water is absorbed into the soil. It is crucial to match the irrigation system's delivery rate with the soil's infiltration rate to avoid excessive horizontal water spread²⁴.

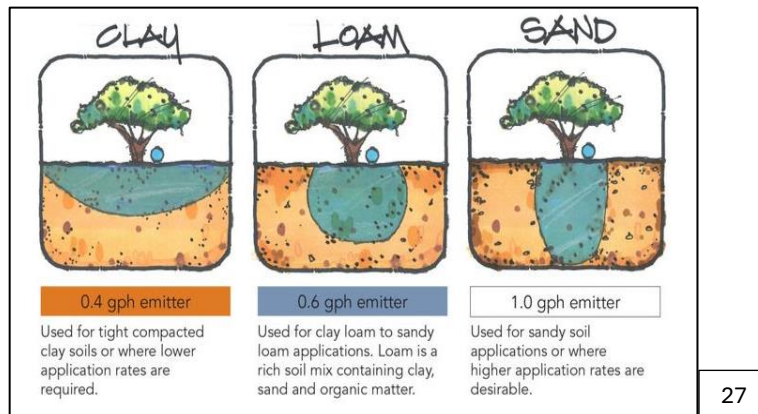
Hydraulic conductivity: This refers to water movement in soil, which is easier in wetter conditions.

Water retention: This is influenced by soil particle size and is the soil's capacity to hold water and nutrients.

Saturated circles: Drip irrigation creates a wetted area beneath the dripper, shaped by soil characteristics and flow rate. Maintaining a balance between water and air in the soil is vital for root oxygenation and water uptake. High flow rates reduce oxygen levels, while lower rates promote better balance. Low delivery rates make the necessary balance possible by achieving good infiltration on moist soils.²⁵



The characteristics of which define soil qualities results in variation on the rate of drip emitters. These characteristics are things such as structure, texture, and soil density. These factors result in different soil watering conditions. For example, when comparing sandy and clay soil types, the sandy types have faster soil infiltration which means it needs to be irrigated more often at a faster rate. On the other hand, clay soil has a high water-holding capacity, resulting in a lower irrigation rate.²⁶



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2.8 Soil types and water needs of the vegetables in the planter box

For the clay-loam type soil that is common in Pu Ngaol, one can estimate that emitter needs to emit around 0.5-0.6gph of water, which is around 2L/Hr in order to balance out soil infiltration rates and ensure an even spread of water²⁸. According to the water needs of the vegetables such a flow rate is not necessary, and due to the fact that the Planter box system is powered by gravity, an even lower drip rate of 0.5-1L/Hr can be used. A study conducted in Nigeria showed that a small efficient and sustainable drip irrigation system fed by gravity that used an emitter with a drip rate of around 0.7L/Hr at a very low pressure with similar soil types could be used.³⁰

Soil texture	Field capacity (mm/mm)	Permanent wilting point (mm/mm)	Available water capacity ¹ (mm/mm)	Infiltration rate (mm/h)
Coarse sand	0.10	0.05	0.05	22
Sand	0.15	0.07	0.08	13
Loamy sand	0.18	0.07	0.11	12
Sandy loam	0.20	0.08	0.12	10
Loam	0.25	0.10	0.15	7
Silt loam	0.30	0.12	0.18	6
Silty clay loam	0.38	0.22	0.16	5
Clay loam	0.40	0.25	0.15	4
Silty clay	0.40	0.27	0.13	3
Clay	0.40	0.28	0.12	2

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2.9 Design considerations for types of emitters used

There are two main categories of emitters: on-line emitters and in-line emitters. These are the design considerations for the emitters used in the planter box.

(A) On-line emitters: These emitters, positioned on lateral pipes, are frequently used in orchards. During installation, the number of drippers, spacing, and location are established and can be modified later³⁰. This design enables customized irrigation for various crops, making it a preferred choice for the planter box proposal.

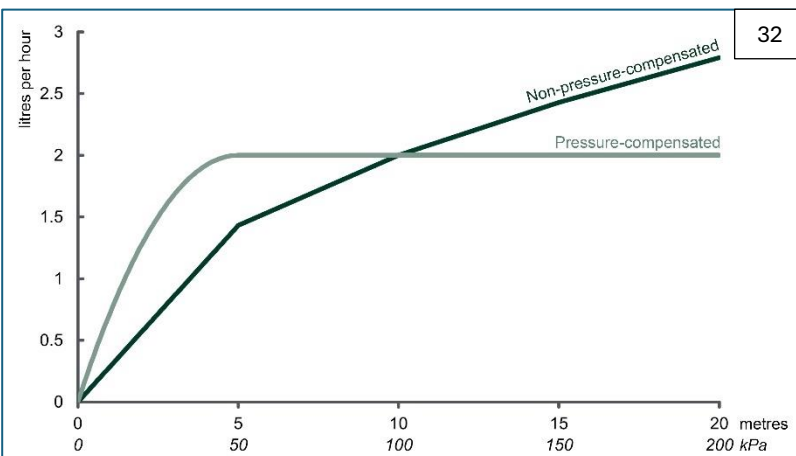
(B) In-line emitters: These form a part of the dripping pipes themselves. They create smooth ducts that allow quick and easy deployment and collection of the driplines. They are used primarily in field crops. These drippers are manufactured in factories with the specific flowrates and intervals already installed. This design is less favourable for the planter box drip system as it is both more expensive, difficult to fix and is not as flexible as the on-line emitter.

There are two main types of on-line emitters in irrigation systems: Long-Path Emitters and Short-Path Emitters.

Long-Path Emitters have extended water paths around a barrel-shaped core, making them larger and suitable for substantial irrigation needs over long distances.

Short-Path Emitters are smaller, more affordable, and ideal for low-pressure systems like gravity flow drip systems from rainwater harvesting. However, they clog easily in mineral-rich water and provide poor water distribution uniformity, making them best for small-scale, cost-sensitive systems³⁰.

Devices known as pressure compensating emitters are able to work with both types. These items can adjust their flow paths based on inlet pressure, ensuring consistent flow rates across various pressures. This feature makes them reliable for uniform water distribution and maintaining efficiency despite pressure changes.



Thus, considering the flow rate needs required by the system, a pressure-compensating short-path emitter is chosen as the emitter which going to be used in the design.

In general, the water requirements of leafy vegetables all have very similar water requirements and rooting depths and thus a one size fits all design for the emitter can be used. The emitter design above³³ is one that could be used for the planter box. According to its specifications, it has a flow rate of around 0.7L/Hr. This design is suitable as it is able to deal with the low pressure of a gravity drip system as it can produce a flow rate the 0.5/Hr. Furthermore, the emitter is designed in such way which reduces the built up of sediments and thus being easier to clean. It can cope with a very large range of water pressures (from 10-80kPa), it can work well with gravity fed planter box system.

3. The Modular Drip Irrigation Planter Box

The main idea of our design is a “Modular Drip Irrigation Planter Box”. The Planter box is a design which makes use of a drip irrigation system as discussed. It is a raised garden bed design, made of wood. The system makes use of a simplified gravity fed drip-irrigation design. Which feeds water at the required pressure for the farm.

For the planter box, certain vegetables have been designated, such as tomatoes, potatoes, and carrots. A specific amount of water is required for these crops. From the estimations of water needed for the planter box, we can start analysing what drip rate and what drip spacing we need. The soil moisture required by these plants is all similar at around 0.35-0.45. This allows for the design of the Planter Box's irrigation system to be a one size fits all, making it easier to grow a diverse selection of food crops.

The main soil type in Pu Ngaol is sandy/clay types. For the Planter Box project clay soil is most desirable. The total area used in the Planter Box is 5 by 3 meters. Resulting in a 21m² area. The rooting depths of these types of vegetables are usually around 45-55cm.

In order to get the mass of soil required for the selected crops, an estimated “bulk density” calculation can be done. An estimate of the total amount of soil needed can be obtained by using assumptions of the water content and density of clay-loam soil, where Volume = 5×3×0.5 = 7.5m³.

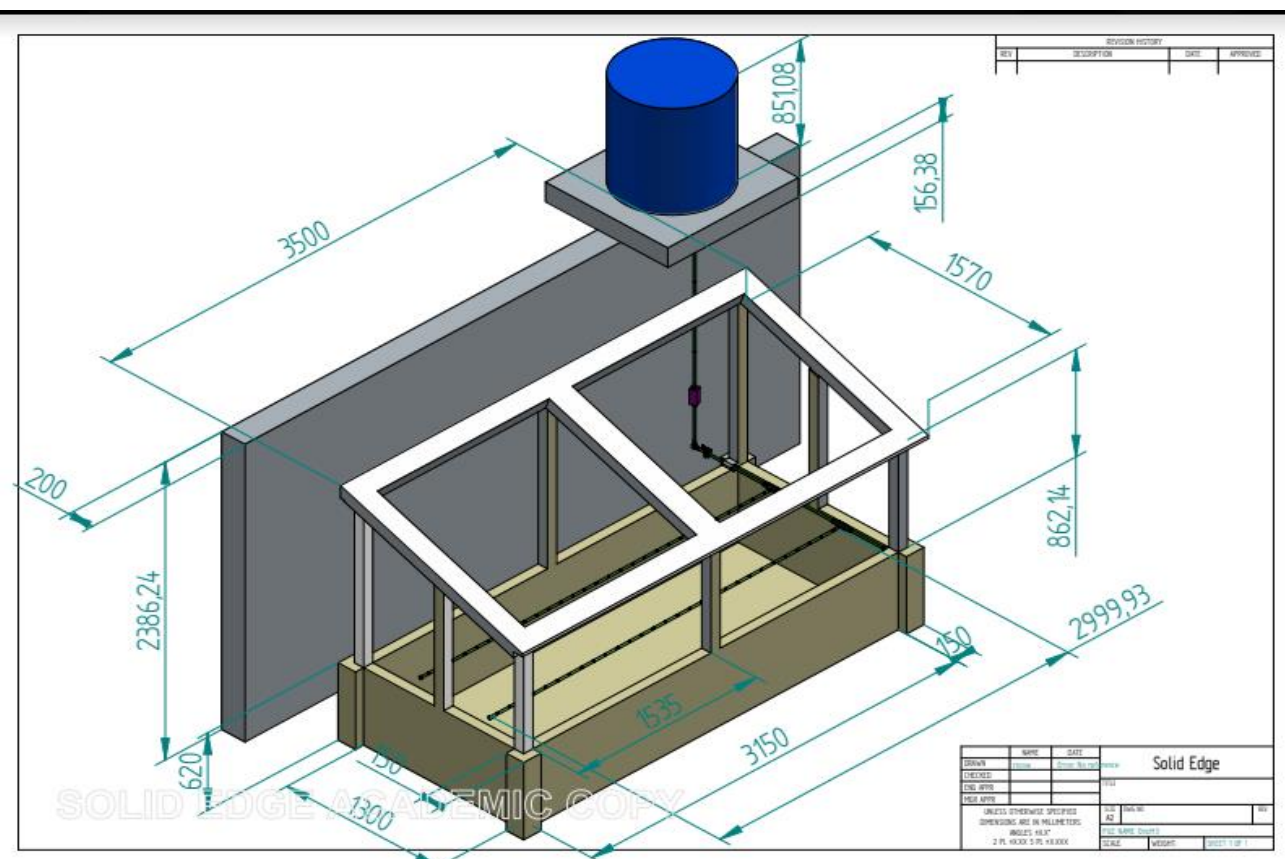
The average density of clay-loam hybrid soil is around 1200kg/m³ when considering average water content of this type of soil.

Thus, using a calculation Density = Mass/Volume, roughly 9000kg of soil is needed.

3.1 Design solution 1

This design is a smaller modular planter box design consisting of a wooden box with discharge holes at the bottom; a small, efficient media filter and a water tank of 850 litres. There is PVC tubing for the main line and sublateral. There is a small greenhouse-like covering for the box, which aims to protect the design from adverse weather conditions and keep the planter box at a constant temperature.

This design aims to use excess rainwater and greywater efficiently in order to develop a sustainable small source of food and vegetables.



3.1.1 The water tank (blue object)

The water tank is 850 litres. The water tank is placed on top of a wall, this would usually be the wall of a residential home. Usually, rainwater from the gutters can keep the tank at a constant volume enough to last throughout the growing season of the crops. This water supply can be supplemented by other forms of grey water, and direct deposits of water into the tank. Due to the water needs of the plants, only around 2-3mm of water a day is needed to supplement their needs. This quantity may grow to 4 mm a day during dry seasons and growth seasons as there is more water evaporation and use.

The material used for the tank is polyethene plastic. Water can fall from the water tank which is usually at the roof height 2.4m (operating head). This means that the water will flow down at a pressure of approximately 20-25 kPa. This ensures the system will have the correct pressure for the water to flow through all the pipes. This can change due to friction in the pipes and other errors. Furthermore, the main pipe is connected straight to the water tank vertically to ensure that there are as few pressure variations in the pipes as possible to ensure that there is always the correct amount of water flowing in the pipes to ensure the maximum amount of water savings.

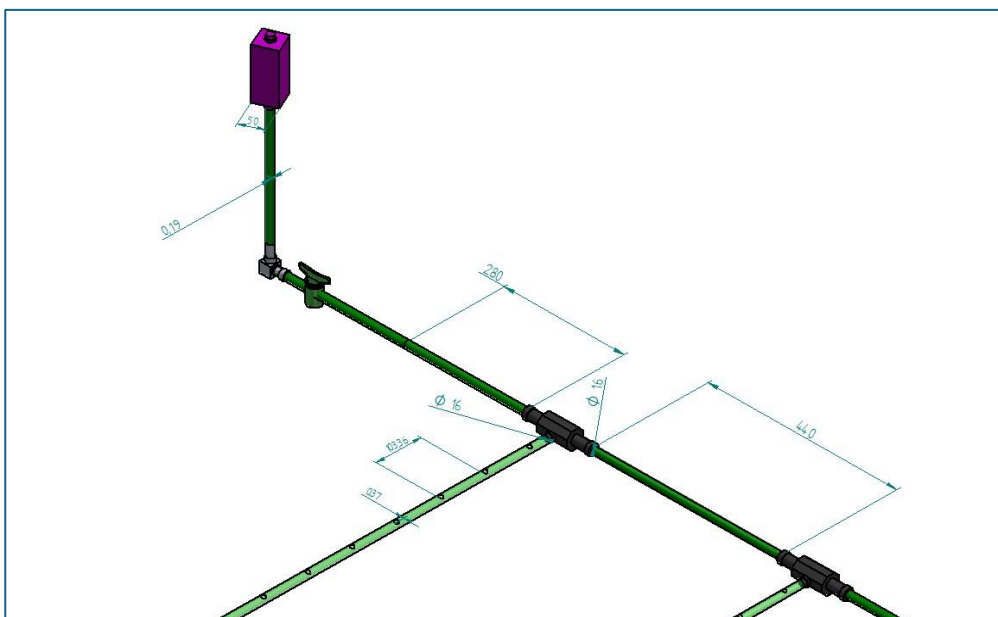
3.1.2 Media filter (purple object)

The filter used is a small general media filter which should be able to deal with removing impurities from general rainwater and grey water. This ensures that the water that flows into the drip irrigation system is clear of sediments and won't harm the crops' health or contaminate the soil.

3.1.3 The covering and the planter box (white and brown object)

There is a greenhouse slanted roof design for the planter box, made of shade cloth. This ensures that in times of heavy rainfall and flooding which occurs often in Cambodia, the plants and soil are protected. The soil won't be eroded away, and the plants can survive and grow. Lastly, the greenhouse ensures that the temperature is kept constant, in terms of drought or colder weather. This results in further water savings and promotes greater plant development. The greenhouse is made of wooden planks and the cloth can be made from general shade cloth. The wooden box is high enough to accommodate good root development and the holes at the bottom of the box flooring allow for the little amount of excess water to easily flow into the ground.

3.1.4 Main pipes and lateral pipes



The piping used is PVC general use plastic type. The main pipe is of diameter 16mm, and the smaller lateral pipes are of diameter 14mm. There is a control valve for the system which can turn the system on and off, this ensures the operation does not waste water when

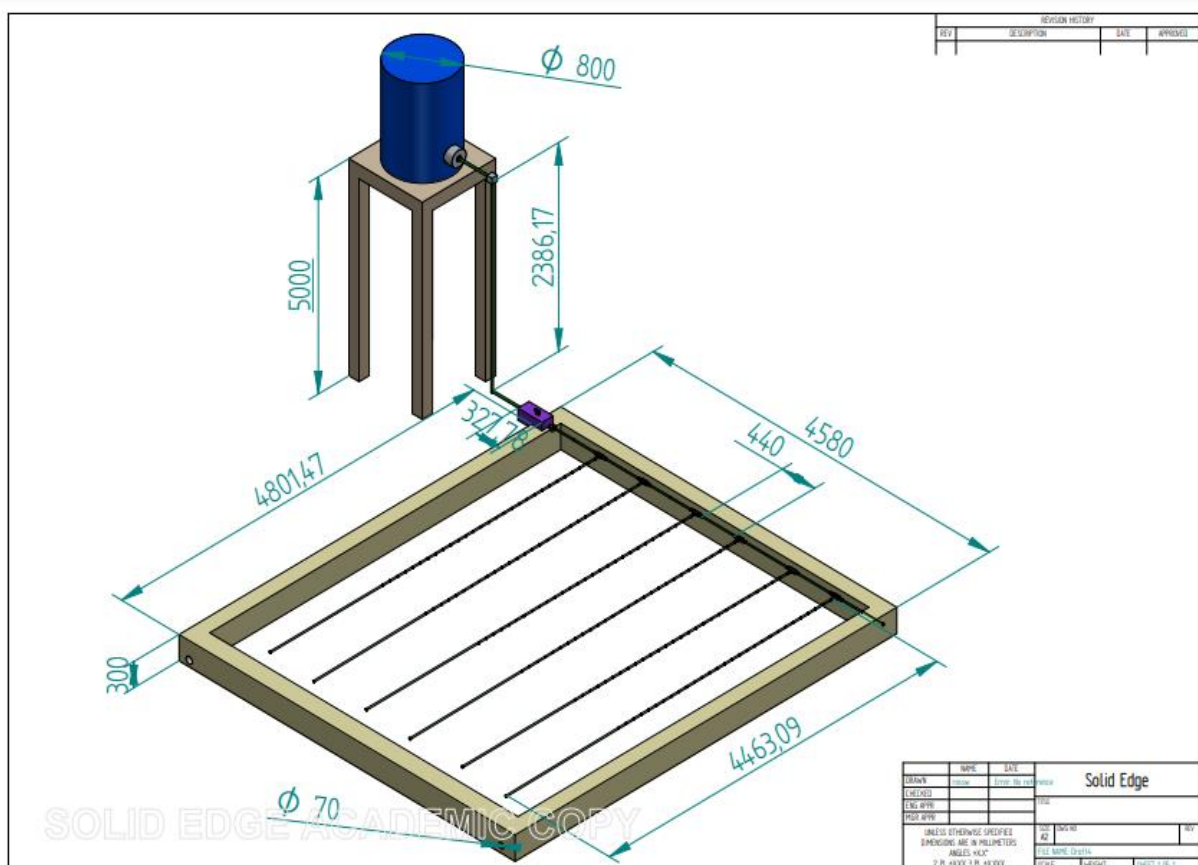
using the system. There are general pipe joints made of plastic which connect the lateral and main pipes together. There are different sized holes in these pipe joints to ensure that the right amount of pressure flows in the system.

There is specific emitter spacing for the design, with it being around 10cm between each emitter hole. The emitter hole itself is very small, being just a diameter of 2mm. There is stop valves at the end of each pipe, which can be removed to easily clean the system and to add more planter boxes as this design is very modular.

Water flows out of the pipes at a flow rate of around 0.3-0.5Lr/Hr, as they are restricted by the specific sized emitter holes. The system irrigates an area of around 4.2m², thus the person operating the system can turn the main controller valve on for at least 3 hours a day. This means the field irrigation rate will always be running at a great level of water saving efficiency.

3.2 Design solution 2

For design solution 2, a much larger and commercial farming approach was considered. This design is a large farm area design which is a raised garden bed design, that is protected on all sides by wooden planks. It consists of a taller water tank tower and a bigger water tank. This design also makes use of a media filter. It has larger pipe diameters of 25mm for the main pipe and 18mm for the smaller lateral pipes. This bigger design focuses on growing a large variety and number of vegetables with a water irrigation system that efficient over a larger area.



3.2.1 The water tank (blue object)

The water tank design is made of polyethylene plastic. It carries a larger volume of 1200 litres. This larger volume is due to the fact that the farm requires more water as there is a larger area (of around 20m²) that needs to be irrigated by the system. To have the required operating pressure for the system, the

water tank is lifted to a height of around 5m. This means that in PVC pipes of diameter 25mm, the operating pressure is around 40-45kPa. Note there is significant drop-off in this pressure due to the fact that the water has to travel longer distances and is affected by more friction in the pipes.

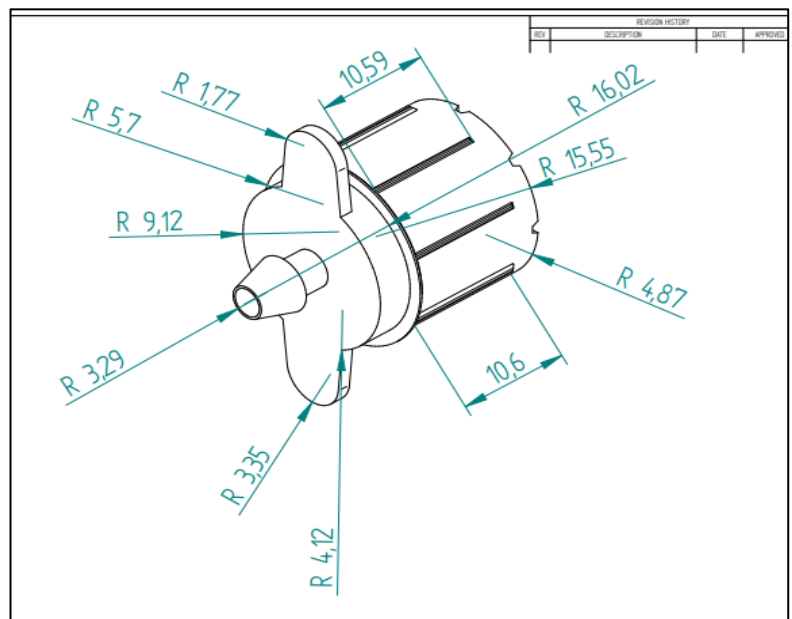
The water tank can be supplied by rainwater and grey water. These sources of water will be mostly enough for the system., but additional cost may occur if the water supply requires more during a dry season, or if other unforeseeable events occur.

3.2.2 Media filter (purple object)

The same type of media filter is used for design 1 and for design 2. For design 2, the filter is of a larger size, so it is able to accommodate the increase volume of the bigger drip irrigation system. (There is a bigger mesh inlet and outlet for water to flow.)

3.2.3 Drip emitter

To counteract the problem of pressure falling off in this design, a type of low pressing compensating emitter is used. It is designed to ensure a consistent water flow, regardless of pressure fluctuations, making it highly efficient for this larger planter box. At the centre of this device is a flexible diaphragm that adjusts according to the water pressure. When the pressure increases, the diaphragm contracts, reducing the flow path to maintain a steady flow rate. Conversely, when the pressure decreases, the diaphragm expands, allowing more water to pass through and ensuring uniform discharge.



3.2.4 The planter box

There are holes in the box, and PVC pipes are installed into them. These pipes help distribute air flow better into the drip system, and in the case of flooding, excess water can be easily removed.

3.2.5 Piping

The piping used is PVC general use plastic type. The main pipe is of diameter 25mm, and the smaller lateral pipes is of diameter 18mm. There is a control valve for the system which can turn the system on and off. This ensures that water is not wasted when using the system.

There are general pipe joints made of plastic which connect the lateral and main pipes together. There are different sized holes in these pipe joints to ensure that just the right amount of pressure flows in the system. There is a specific emitter spacing for the design, with it being around 20cm between each emitter hole. The emitter hole itself is very small, being just a diameter of 8mm. There is stop valves at the end of each pipe, which can be removed to easily clean the system. Water flows out of the pipes at a flow rate of around 0.2-0.3Lr/Hr. The area irrigated is rather large and thus the system needs to be turned on for around 5-6 hours a day to ensure that the crops receive the correct amount of water each day.

3.3 Analysis of designs

	Design solution 1		Design solution 2	
Criteria	Score	Justification	Score	Justification
<u>Sustainability:</u> (How does the design promote sustainability, is it eco-friendly, what are the materials made of. Can the design coexist with the natural environment without affecting it detrimentally?)	4	The pipes and components are made from a high-density polyethylene which is lightweight and durable. Therefore, they require minimal service during their life. The design allows for easy replacement of parts. The greenhouse part ensures sustainability by saving on water through reduced evaporation as well as improving humidity levels within the system.	3	The materials used in this system are selected for robustness and durability, with a focus on minimizing environmental impact. The water tank and emitters may be made from recycled plastics or other sustainable materials that can withstand the demands of extensive agricultural use.
<u>Water conservation:</u> (Is the design able to effectively conserve water, does the system implement the drip irrigation design efficiently.)	4	The design allows for precise water delivery directly to the plant roots, minimizing loss through evaporation and runoff. The greenhouse component enhances this efficiency by creating an environment that reduces water evaporation and protects crops from harsh weather conditions. The use of PVC results in minimal leakage and efficient water flow, further promoting water conservation. The inclusion of a large water tank for rainwater collection provides a sustainable source of water for use and in droughts.	4	The larger scale irrigation system with a pressure-compensating emitter works for extensive agricultural use and does well in managing water over large areas. The system's pressure-compensating emitters maintain a consistent flow rate regardless of pressure variations, ensuring even water distribution and minimizing waste of water. There is a reduced reliance on external water supplies.
<u>Complexity:</u> (How difficult is the design to use and setup. How much materials. Is the design intuitive?)	5	The design is relatively simple in terms of complexity. Its design focuses on providing targeted irrigation for planter boxes and small-scale gardening setups. The system's components, including lightweight tubing and efficient connectors, they are straightforward to install and maintain. The greenhouse addition, while enhancing water conservation, does not significantly increase the system's complexity. This simplicity makes it accessible and user-friendly, ideal for those with limited technical expertise.	2	Design 2, because of its greater scope and sophisticated features is rather complex. The system has complicated pressure-compensating emitters that need to be calibrated precisely in order to guarantee a constant water flow Furthermore the use of a large water tank and tower for the collection and distribution of rainwater results in more upkeep and thought to design.

Emitter and pipe usage: (How effective is the layout of the lateral and main pipes in the design, are the emitters used efficient?)	4	Design 1 is a straightforward design in terms of emitter and pipe usage. This system uses lightweight, flexible tubing and simple drip emitters that are easy to install and replace.	4	Design 2 employs a more complex arrangement of emitters and pipes to manage water distribution over extensive agricultural areas. The system uses advanced pressure-compensating pipes and a network of larger-diameter pipes designed to handle higher volumes of water over longer distances.
Economic cost: (How expensive is the design overall. How much does the design cost to create and use?)	5	Design 1 is economically advantageous for subsistence farming. The initial cost is low due to the use of simple components, which are inexpensive to produce and purchase. Installation and maintenance costs are also minimal, as the system's straightforward design requires no specialized tools or expertise. The greenhouse component, although adding a slight increase to the initial investment, contributes to long-term savings by improving water efficiency.	3	Because of Designs 2's complexity and scope of use, larger-scale irrigation systems come at a higher financial expense. The cost of manufacturing and installing the large diameter pipes and advanced emitters needed, is higher. Furthermore, the original expenditure is increased when a water tank tower for rainwater collection is included. The system's effectiveness and long-term advantages, such as decreased water waste and increased agricultural yields, justifies the higher upfront expenses for large-scale agricultural operations.
Total:	22		16	

3.3.1 Summary

When comparing the two drip irrigation systems, it is clear that each has its own advantages suitable for different situations.

The small modular system with an attached greenhouse is designed to be simple, inexpensive, and easy to use (design 1), therefore it is perfect for small-scale cultivation. It is made up of light hoses, uncomplicated water release points, and low upkeep needs, which means it can be operated effectively with little financial outlay.

On the other hand, a larger scale irrigation system employing pressure compensating emitters and wider coverage demands more upfront investment but provides greater efficiency and productivity gains especially for commercial farming designs. (design 2).

Through comparing both designed, Design 1 has proven to be the optimal design option for combatting the problem of food insecurity in the village of Pu Ngaol.

4. Financial Analysis

	Item	Size/length	Quantity	Unit cost (\$)	Total cost (\$)
	Water tank	850	1	108.83	108.83
	Tank stand	1x1	1	110.3	110.3
	Main line pipe	1400m, D16	1	23.13	23.13
	Lateral line pipe	3000mm, D14	2	32.07	64.14
	Hole emitter	D10	60	2.76	165.6
	Control valve	Standard	1	35.95	35.95
	Elbow joints	D16, Inner D14	2	8.35	16.7
	T joints	D16, Inner D14	2	6.56	13.12
	End caps	2/3 Inch	3	4.40	13.2
	Sand Filter	Standard	1	10.00	10.00
	Wooden Box	3x1.6	1	30.00	30.00
	Greenhouse Stand planks	standard	1	16.33	16.33
	Shade Cloth	3m ²	1	2.06	6.20
	Total				613.5

The initial investment for the drip irrigated planter box design involved significant material and equipment costs. To minimize shipping expenses, essential materials such as the water tank, tank stand, pipes, and fittings were procured locally, totalling \$613.5. Equipment costs included components of the drip irrigation system, greenhouse parts, and assembly tools, although these specific costs were integrated within the overall material expenses. Local labour was employed to assemble the components and construct the planter boxes. The training cost can be occurred for the training of the villagers to understand how to use the equipment. Therefore, the total initial investment cost, combining materials and labour, came to \$658.5.

Operating costs were calculated based on water consumption, maintenance, and potential soil and fertilizer use. Water costs, although not specified, were considered in the operational analysis. Regular maintenance, essential for ensuring the longevity of wooden components and the drip irrigation system, was estimated at \$52 annually. This takes into account possible breakage of materials used in the design and the cost of replacing them.

Income generation for the project relied on sales revenue from the planter box modules and additional revenue streams. The income generated can be estimated from the produce of vegetables, such as tomatoes, carrots, and potatoes. With an estimated selling price of \$165 per module and an annual sales projection of around \$90 with accounting for the produce generated from other modular planter boxes. Additional income opportunities, such as offering maintenance services and training courses on

drip irrigation and organic farming, were anticipated to bring in another \$55 annually. Consequently, the total annual revenue was estimated at \$145.

To determine the break-even point, both fixed and variable costs were considered. It was calculated that selling approximately 5 modules would cover the initial investment. This means producing at least 9 modular planter boxes which can be used for the village allows them to generate some form of profit allowing the villagers to pay back the initial investment for the project. Given the annual sales estimate of 5 modules, the project was expected to reach the break-even point in approximately 2 years.

Initial investment to fund the project would be procured through fundraising events and campaigns, as well as securing funding from portions of taxes as these planter boxes will be used to better the health and food security of Pu Ngaol.

In conclusion, the financial analysis demonstrated that the drip irrigated planter box project was both viable and profitable in rural Cambodia. By leveraging local resources for materials and labour, and exploring diverse revenue streams, the project not only ensured financial sustainability but also offered significant benefits to the local community. Strategic planning and ongoing market research were emphasized as critical to maximizing the project's financial performance and community impact.

5. Conclusion

The implementation of planter boxes equipped with drip irrigation systems will significantly support the community of Pu Ngaol in their subsistence farming efforts, thereby addressing their food shortage issues.

Drip irrigation emerges as the optimal choice for the cultivation of small crops intended for subsistence. This system is less intensive and ensures a more efficient distribution of water compared to alternative irrigation methods. It promotes water conservation while simultaneously boosting crop productivity. The chosen filtration method, a sand filter, is ideal given that the primary water source will be grey water. Unlike more complex filters, such as green sand filters, which are unnecessary in this context, the sand filter efficiently removes impurities without the need to address minerals like iron, which are not present in grey water.

The system we have developed employs a planter box integrated with a network of pipes designed for drip irrigation. This setup includes a water tank and a sand filter, ensuring the delivery of high-quality water to the plants. An emitter, strategically positioned within the pipes, regulates both the volume and pressure of the water, providing precise irrigation tailored to the plants' needs.

This configuration is highly effective in delivering optimal hydration to the plants while maintaining water quality. Its straightforward design facilitates easy construction and assembly, making it an accessible solution for various users. Additionally, the design is economically viable, with readily available materials such as PVC pipes, further enhancing its practicality and affordability.

Furthermore, the cost-effectiveness of the drip irrigation system is noteworthy, especially when compared to more complex and expensive systems like sprinklers. The ease of setup and maintenance of this system makes it particularly suitable for the local context of Pu Ngaol.

The irrigation system will incorporate advanced control heads, which are valves attached to the pipes to regulate water flow precisely. To ensure simplicity and ease of use for the villagers, water pumps will facilitate a gravity-fed system. The primary distribution network will consist of PVC piping, chosen for its durability and accessibility.

Emitters, strategically placed near the soil, will ensure that the plants receive an efficient and adequate amount of moisture.

The use of planter boxes is ideally suited for the cultivation of these vegetables. These structures provide an optimal growing environment, allowing for efficient water management and protection against extreme weather conditions. Thus, planter boxes present a practical and effective solution for both subsistence farming and small-scale commercial agriculture in Cambodian communities.

In conclusion, the drip irrigation system with refillable water capabilities presents a comprehensive solution for the agricultural challenges faced by the people of Pu Ngaol. This system not only conserves water but also enhances food availability, thereby exerting a positive impact on the community's agricultural productivity and overall well-being. The villagers will acquire a valuable new skillset, enabling them to utilize this advanced agricultural method sustainably over the long term. Additionally, they will have the opportunity to generate income from the sale of the vegetables produced, enhancing their economic stability.

Overall, this initiative represents a cost-effective strategy to significantly improve agricultural practices in Pu Ngaol, fostering both community development and self-sufficiency.

6. References

- [1] file:///C:/Users/rossw/Downloads/2023-24-Engineering-for-People-Design_Brief_Cambodia%20(2).pdf
- [2] weatherandclimate.com. (n.d.). Phumi Pu Ngai (1), Mondol Kiri, KH Climate Zone, Monthly Averages, Historical Weather Data. [online] Available at: [https://weatherandclimate.com/cambodia/mondol-kiri/phumi-pu-ngai-1#:~:text=Phumi%20Pu%20Ngal%20\(1\)%20Climate%20Summary&text=The%20district](https://weatherandclimate.com/cambodia/mondol-kiri/phumi-pu-ngai-1#:~:text=Phumi%20Pu%20Ngal%20(1)%20Climate%20Summary&text=The%20district).
- [3] University of Rhode Island (n.d.). Drip Irrigation. [online] The University of Rhode Island. Available at: <https://web.uri.edu/safewater/protecting-water-quality-at-home/sustainable-landscaping/drip-irrigation/#:~:text=Unlike%20other%20forms%20of%20irrigation>.
- [4] www.netafim.co.za. (n.d.). Drip irrigation solution. [online] Available at: <https://www.netafim.co.za/drip-irrigation/#:~:text=With%2095%20%2D%20100%25%20water%2D> [Accessed 22 May 2024].
- [5] Bhattarai, Surya & Palada, Manuel & Midmore, David & Wu, D. & Salas, Rosario. (2011). On-farm evaluation of fertilizer briquettes and low-cost drip irrigation for smallholder vegetable production in Cambodia. *Irrigation and Drainage*. 60. 318 - 329. 10.1002/ird.571.
- [6] AGRIVI (2022). Drip Irrigation: Full Guide | Function | Parts - AGRIVI. [online] AGRIVI. Available at: https://www.agrivi.com/blog/drip-irrigation-as-the-most-efficient-irrigation-system-type/#The_Cons_of_a_Drip_Irrigation_System [Accessed 22 May 2024].
- [7] IRRIGATION VALDUCCI. (n.d.). Drip irrigation: all you need to know. [online] Available at: <https://irrigationvalducci.com/en/drip-irrigation-all-you-need-to-know>.
- [8] Anon, (2023). Water conservation in sustainable agriculture. [online] Available at: <https://tracextech.com/water-conservation-in-sustainable-agriculture/>.
- [9] Changkun Guan, Xianlei Ma, Xiaoping Shi, The impact of collective and individual drip irrigation systems on fertilizer use intensity and land productivity: Evidence from rural Xinjiang, China, *Water Resources and Economics*, Volume 38, 2022, 100196, ISSN 2212-4284, <https://doi.org/10.1016/j.wre.2022.100196>.
- [10] Aziz, Marjan & Rizvi, Sultan & Iqbal, Muhammad & Syed, Sairah & Ashraf, Muhammad & Anwer, Saira & Usman, Muhammad & Tahir, Nazia & Khan, Azra & Asghar, Sana & Akhtar, Jamil & Punjab, Pakistan. (2021). A Sustainable Irrigation System for Small Landholdings of Rainfed Punjab, Pakistan A Sustainable Irrigation System for Small Landholdings of Rainfed. *Sustainability*. 13. 10.3390/su132011178.
- [11] Yang, P., Wu, L., Cheng, M., Fan, J., Li, S., Wang, H. and Qian, L. (2023). Review on Drip Irrigation: Impact on Crop Yield, Quality, and Water Productivity in China. *Water*, [online] 15(9), p.1733. doi:<https://doi.org/10.3390/w15091733>.
- [12] Gupta, P. (1970). Effect of drip irrigation on growth and yield of onion (*Allium cepa* L.). *Journal of Spices and Aromatic Crops*. 32.10.25081/josac.2018.v27.i1.1012
- [13] Manda, Raghavendra & Addanki, Venkata Avinash & Srivastava, Dr. Seweta. (2021). ROLE OF DRIP IRRIGATION IN PLANT HEALTH MANAGEMENT, ITS IMPORTANCE AND MAINTENANCE. *Plant Archives*. 21. 1294 - 1302. 10.51470/PLANTARCHIVES.2021.v21.S1.204.

- [14] BYJUS. (n.d.). What are the pros and cons of drip irrigation? Answer at BYJU'S IAS. [online] Available at: <https://byjus.com/ias-questions/what-are-the-disadvantages-of-drip-irrigation/>.
- [15] <https://wikifarmer.com/components-and-types-of-drip-irrigation-systems/>
- [16] Energypedia.info. (2019). Drip Irrigation - energypedia.info. [online] Available at: https://energypedia.info/wiki/Drip_Irrigation.
- [17] Dewidar, Ahmad & Al-Fehaid, Youssef & Al-Hilal, Sabry & Ben Salah, Mohamed. (2016). Water saving in arid regions: A comparison of surface and subsurface drip irrigation systems. American Journal of Innovative Research and Applied Sciences. 289-296.
- [18] edis.ifas.ufl.edu. (n.d.). AE539/AE539: Basic Tips for Designing Efficient Irrigation Systems. [online] Available at: <https://edis.ifas.ufl.edu/publication/AE539>.
- [19] Tarafdar, J.. (2023). Re: Can hard water affect crops?. Retrieved from: https://www.researchgate.net/post/Can_hard_water_affect_crops/63d794f74a5264e86b0e4bfc/citation/download.
- [20] Department of Jobs, P. and R. (2023). *Filtration for drip irrigation - Agriculture*. [online] Agriculture Victoria. Available at: <https://agriculture.vic.gov.au/farm-management/water/irrigation/drip-irrigation/filtration-for-drip-irrigation>.
- [21] De Souza, F. H., Pizzolatti, B. S., & Sens, M. L. (2021). Backwash as a simple operational alternative for small-scale slow sand filters: From conception to the current state of the art. Journal of Water Process Engineering, 40, 101864. <https://doi.org/10.1016/j.jwpe.2020.101864>
- [22] Standard bricks Concrete tiles -whole tiles are set on quarter- tiles as illustrated or of * concrete tiles * household bricks * large gravel (40 -100mm). (n.d.). Available at: <https://www.lboro.ac.uk/media/wwwlboroacuk/external/content/research/wedc/pdfs/technicalbriefs/15.%20Slow%20sand%20filter%20design.pdf>.
- [23] Low Cost Filtration of Domestic Wastewater for Irrigation Purpose - Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/Slow-sand-filter-FS_fig2_326651670 [accessed 24 May, 2024]
- [24] www.netafim.co.za. (n.d.). Dripper spacing and delivery rate - Understanding water distribution... [online] Available at: <https://www.netafim.co.za/blog/dripper-spacing-and-delivery-rate---understanding-water-distribution-in-soil/> [Accessed 22 May 2024]. Note photograph also comes from this source
- [25] www.fao.org. (n.d.). CHAPTER 6. DRIP IRRIGATION. [online] Available at: <https://www.fao.org/4/s8684e/s8684e07.htm#6.2%20drip%20system%20layout> [Accessed 22 May 2024].
- [26] Palada, Manuel & Bhattarai, Surya & Wu, Deng-lin & Roberts, Michael & Bhattarai, Madhusudan & Kimsan, Ros & Midmore, David. (2011). More Crop Per Drop: Using Simple Drip Irrigation Systems for Small-scale Vegetable Production.
- [27] Pinterest. (n.d.). Why Use Drip Irrigation? | Irrigation, Drip irrigation, Sandy soil. [online] Available at: https://ar.pinterest.com/pin/763149099338363796/?amp_client_id=CLIENT_ID%28_%29&mweb_unauth_id=&simplified=true [Accessed 22 May 2024].
- [28] White, Peter & Oberthur, Thomas & Dobermann, A. & Ros, Casana. (2000). Classifying soils for agronomists in Cambodia I. The Cambodian agronomic soil classification system (CASC). Soil Use Manage. 16. 12-20.
- [29]] Oke, Mary & Ukume, Josphine & Aremu, Tolulope & Osilaechuu, Adaosila & Iwar, Raphael. (2019). Design of a Gravity Drip Irrigation System for Small Scale Vegetable Gardens. 61-72.
- [30] Vince, V.B. (2022). The Complete Guide to Drip Irrigation Emitter Types. [online] Hills Irrigation - Artists with Water. Available at: <https://www.hillsirrigation.com.au/drip-irrigation-emitters-types/>
- [31] Price, M. (2019). Irrigation Fundamentals. [online] Water Well Journal. Available at: <https://waterwelljournal.com/irrigation-fundamentals-10/> [Accessed 22 May 2024].
- [32] Department of Jobs, P. and R. (2023). Planning a drip irrigation system - Agriculture. [online] Agriculture Victoria. Available at: <https://agriculture.vic.gov.au/farm-management/water/irrigation/drip-irrigation/planning-a-drip-irrigation-system>.
- [33] store.rainbird.com. (n.d.). FE40-25S - Drip Flag Emitter, 4 GPH (pack of 25). [online] Available at: <https://store.rainbird.com/fe40-25s-drip-flag-emitter-4-gph-pack-of-25.html> [Accessed 22 May 2024].

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